WEB READINESS FOR AI AGENTS

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ABSTRACT

Autonomous AI agents need the web to support both information retrieval and automated action. This study evaluates public web readiness by analyzing 50,000 domains sampled across the top one million global websites from the research-backed Tranco popularity ranking through 61 automated checks using direct HTTP requests without logins or client-side JavaScript. Basic homepage text was accessible on 45.71% of domains without requiring a browser to run scripts. When evaluating whether sites provide readable content while simultaneously allowing automated access, on 30.16% of all domains, the text rule passed and both requests using the crawler names GPTBot and PerplexityBot were recorded as HTTP 200 ("OK") without being blocked by anti-bot firewalls. Established discovery signals were relatively common, with sitemaps and robots-file markers and machine-readable page metadata (JSON-LD) passed on 27.37% and 21.37%, respectively. Dedicated agent infrastructure remains rare, as at least one check for specialized guidance files, Markdown negotiation, or Model Context Protocol (MCP) endpoints passed on 7.50% of domains, and candidate API descriptions passed on 0.16% with an OpenAPI version marker. While agents can retrieve static text from parts of the web, structured public interfaces for autonomous action remain rare.

Keywords: AI agents, agent readiness, web measurement, robots.txt, llms.txt, Model Context Protocol, OpenAPI, Tranco, CrUX

1 INTRODUCTION

AI agents are software systems in which an AI model chooses actions and uses tools to work toward a user's goal. Those tools can include a web browser, a search service, or an application programming interface (API), which lets software request data or operations from another service. An agent can use the result of one action to decide what to do next [31]. For example, it may search for product information, compare alternatives, or complete an online form.

In July 2025, OpenAI introduced ChatGPT agent with tools for browsing websites and using APIs [32]. Such products make website readiness a practical question for users and developers. A user can ask an agent to complete a task, but the agent still needs to find the right information and a way to use the relevant service.

For an AI agent to execute tasks autonomously, a website must support two distinct capabilities, information retrieval and service execution. Today, these capabilities are split. Human-facing websites often publish rich informational text but lack machine-readable API specifications, forcing agents to rely on expensive visual browsing or fragile HTML scrapers. Conversely, software services frequently provide structured APIs but hide them behind authentication or omit machine-discoverable documentation from their public homepages.

Because full browser emulation carries high computational latency and cost, many agent architectures rely on lightweight, direct HTTP requests to locate resources. For an agent to gather information, a website must satisfy two basic conditions, it must return readable text directly without requiring browser JavaScript execution, and its network firewalls must not block automated requests. While 45.71% of domains returned readable HTML text, only 30.16% simultaneously allowed test requests carrying common AI agent identifiers (GPTBot and PerplexityBot) through without blocking them. This raises a fundamental empirical question of how much of the existing public

web an autonomous agent can understand and use through standard HTTP requests alone, without relying on client-side JavaScript execution or authenticated credentials.

This analysis asks how often the selected rules match, how their matches overlap, and how the results depend on site rank and collection procedure. It combines complete check counts with rank-adjusted estimates, paired repeat-scan comparisons, and tests with prepared responses. The contribution is a reproducible description of these public-web signals and the limits on interpreting them as readiness; completed agent tasks were not part of the scan.

2 Related Work

Research on AI agents usually tests whether models can finish specific tasks. For example, WebArena measures whether an agent can book a flight, post on a forum, or buy an item on simulated websites [33]. Those benchmarks test the model's problem-solving skills inside controlled environments. In contrast, this study does not measure model skill or task completion. It checks whether real, public websites provide the basic text, files, and interfaces that an agent needs to work autonomously.

Existing web measurement projects analyze frontend technologies and performance, but they do not examine whether sites support automated software. For example, the HTTP Archive tracks CMS platforms, JavaScript libraries, page weight, and loading times designed for human browsing [21]. Studies looking at newer agent formats show that a successful server reply does not mean the file is valid. When Thunderbit looked for llms.txt files across 10,000 domains, 1,606 addresses returned an HTTP 200 code, but only 586 actually contained valid files [22]. The other servers returned ordinary HTML web pages, error messages, or redirects. An agent cannot assume a file exists just because an address responds.

Other studies inspect robots.txt files to see what websites allow automated crawlers to do [23; 24; 25]. However, what a website writes in robots.txt does not guarantee what its servers allow in practice. Security systems and firewalls often block automated visitors even when robots.txt allows them. For this reason, this study checks both the written rules in robots.txt and sends test requests using crawler names such as GPTBot and PerplexityBot to see if the server lets them through.

Publishing these files also does not mean agents use them. An analysis by Ahrefs of 137,210 domains found that 97% of detected llms.txt files received no requests at all in May 2026 [26]. In commercial tooling, Ora's methodology page lists 126 checks across four groups as of October 2026 [29]. Because different studies use different websites and scoring rules, their numbers cannot be compared directly. This study uses a clear, open set of 61 automated checks to measure basic access, resource discovery, and service interfaces across 50,000 domains.

3 Study Design

Evaluating agent readiness requires looking at how automated software actually interacts with web infrastructure. This study organizes its 61 checks into three primary operational areas. Access checks verify whether homepages return standard response codes and readable text without requiring browser rendering. Discovery checks scan for sitemaps, crawler permissions, and machine guidance files that tell agents where resources reside. Service interface checks probe for structured API specifications, protocol endpoints, and authentication guidelines needed for execution.

Because different agents require different tools, a single pass mark cannot describe readiness. A research agent searching for articles only needs readable text and clear sitemaps, while a purchasing agent requires structured API endpoints and transaction support; optional files such as llms.txt are not requirements for all tasks. This study records each signal independently, establishing where public web infrastructure supports autonomous software and where gaps remain.

3.1 Sample

The sampling frame comes from the Tranco ranking, an established scientific benchmark that measures website popularity by combining multiple web traffic providers (such as Cisco Umbrella, Alexa, and Majestic) over a 30-day window to filter out transient spikes and manipulation [1; 27; 28].

This study uses the top one million global domains from Tranco list ZJQ6G, generated on 2 September 2026.

The sample includes every domain ranked 1 through 10,000. From each of the next two rank groups, it includes 20,000 domains picked at random, with no domain picked twice. The random seed is 20260903, allowing the same selection to be reproduced.

Table 1: Rank groups in the sampling frame and study sample.

Rank range	Domains in full list	Domains sampled
1-10,000	10,000	10,000
10,001-100,000	90,000	20,000
100,001-1,000,000	900,000	20,000

Each listed hostname is one observation. Several names can lead to the same website, so the counts do not represent unique organizations or website owners. The unadjusted percentages describe this sample; Section 4.3.1 gives estimates adjusted to the rank mix of the full list.

The study also checks whether each domain appears in the July 2026 Chrome User Experience Report (CrUX) top-million list of web addresses [2; 3]. For this comparison, it ignores the URL protocol and port number and treats www.example.com and example.com as the same hostname. It finds 23,592 matches. CrUX groups addresses into broad popularity bands and applies rules about popularity and being findable on the public web. CrUX is also one of the sources used to build Tranco [28].

3.2 Why these 61 checks

The checklist inspects public pages, standard files, and HTTP response headers. These signals reflect what automated agents need at different stages of a task [31]. Basic access tests verify whether pages return readable text. Discovery checks identify sitemaps, crawler permissions, and machine guidance files. Interface checks look for API descriptions, protocol endpoints, and authentication rules. General metadata and security headers provide operational context, while payment signals matter for commercial interactions.

All checks evaluate public URLs and server responses without logging into user accounts, running browser JavaScript, filling forms, or completing transactions. This provides an objective baseline for comparing websites across the web, though it does not test private or interactive workflows. The checklist originated from merging an early AI-standards list with general web quality checks, later expanding to cover AI search, autonomous agents, and structured metadata.

The 61 checks represent an exploratory inventory rather than a definitive standard for agent readiness. Grouping choices affect the total count; for example, verifying an MCP server and checking its capabilities count as two separate checks, whereas testing two crawler user-agents produces one combined check. Table 2 outlines the six categories.

Table 2: Purposes and sizes of the six checklist categories.

Category	Purpose	Checks
Discovery	Find resource lists, crawler rules, and guidance files	13
Access	Check response codes, available text, formats, and response headers	9
Interface and MCP	Look for possible service addresses, API descriptions, and access-permission information	10
Security	Look for security headers and contact details, without testing an agent's permission to act	11
SEO and structured data	Find titles, language tags, and descriptions of page content used by search engines and other software	15
Payments	Look for payment markers and terms relevant to commercial tasks	3

The checklist includes established web standards such as sitemaps and OpenAPI [5; 10], recent proposals such as llms.txt [6], and experimental endpoint probes. While these checks capture useful signals, passing more checks does not prove that an AI agent will succeed at a given task.

3.2.1 Requests and matching rules

The scanner implementation begins with 34 fetch requests per domain and can check up to six additional resources if initial probes succeed. Following HTTP redirects can increase this request count. The probes target the homepage over HTTPS alongside well-known paths such as /robots.txt, /llms.txt, and /openapi.json, with a 4.5-second timeout per operation. One homepage request requests Markdown or plain text. Two other requests test User-Agent headers for GPTBot and PerplexityBot. The scanner does not run client-side JavaScript or log in. Appendix B provides the complete matching rules.

The text check evaluates the standard homepage response, falling back to the Markdown response if the standard page is empty. It strips scripts, stylesheets, and HTML tags, requiring more than 300 characters of text to pass. Error pages and hidden text can still satisfy this count.

The crawler check records whether both, one, or neither of the test requests return an HTTP 200 response. The MCP check tests two candidate service paths with a JSON-RPC initialization payload, passing if the server returns an HTTP status between 200 and 299 with the string jsonrpc. A second check parses the response as JSON and looks for an explicit capabilities object.

3.3 Saved results and repeat scans

Each saved domain record contains 61 check labels, an overall score, category scores, the scan timestamp, and the duration in milliseconds. The four possible labels are pass, warning, fail, and not applicable. A warning indicates a partial match or an unverified signal, while not applicable marks an optional feature that was absent or dependent on another prerequisite check. Appendix C explains the scoring formulas and rank weighting.

The dataset contains one record for each of the 50,000 sampled domains, collected over 18.77 hours on September 3-4, 2026 UTC. To address transient network drops, the collection pipeline automatically rescanned 21,547 domains whose initial score was 10 or below, replacing earlier records with new results.

3.4 Tests with prepared responses

To test the scanner's matching rules, the implementation was evaluated against 13 prepared test cases with known responses, including empty replies, server errors, and standard HTML pages. Because the true inputs were known, these tests revealed where heuristic rules can produce false passes or misinterpret server errors.

3.5 Comparisons

The analysis investigates several key patterns across the 50,000 domains.

First, it examines the joint adoption of three emerging agent technologies, llms.txt files, Markdown content negotiation, and Model Context Protocol (MCP) endpoints, counting how often domains implement none, one, two, or all three standards.

Second, it evaluates practical access by measuring how often domains return readable homepage text while simultaneously allowing automated crawlers through with HTTP 200 responses.

Third, it compares domains listed in Chrome's User Experience Report (CrUX) against unlisted domains, standardizing both groups to identical rank distributions to determine whether higher pass rates reflect site popularity or structural differences.

Finally, it compares paired repeat scans among low-scoring domains to determine how network retries affect overall scores.

4 Readiness Results

4.1 Infrastructure indicators

All 50,000 sampled domains have a complete record across the 61 checks. The overall checklist score averages 25.7 out of 100, with a median of 22. Table 3 presents results for eleven indicators spanning the six categories across all 50,000 domains. Appendix A provides complete counts for all 61 checks.

Table 3: Selected check results across all 50,000 domains.

Check	Pass count	Pass share
robots.txt rule check	21,881	43.76%
llms.txt check	3,310	6.62%
Markdown-like response	1,474	2.95%
HTTP 200 for both test requests	18,912	37.82%
MCP response marker	910	1.82%
OpenAPI version marker	78	0.16%
security.txt contact marker	2,288	4.58%
JSON-LD script check	10,687	21.37%
Sitemap and robots.txt markers	13,686	27.37%
index.md body check	5,337	10.67%
Payment-related markers	163	0.33%

Both crawler-named requests were recorded as HTTP 200 on 18,912 domains (37.82%), one on 3,901 (7.80%), and neither on 27,187 (54.37%). The final category combines connection timeouts, DNS failures, and non-200 HTTP responses.

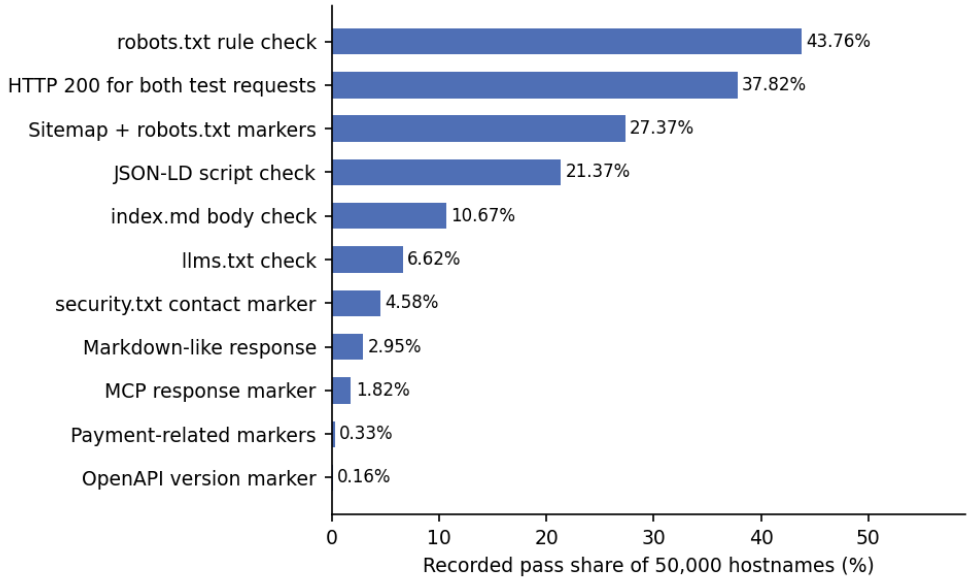


Figure 1: Percentage of all 50,000 sampled domains passing each of eleven checks.

4.1.1 How often the three selected checks pass together

At least one of the three selected checks passes on 3,748 domains (7.50%), while all three pass on 833. Most domains with any pass have only one of the three, but most MCP matches occur in the group passing all three. Table 4 counts each domain once.

Table 4: Number of selected checks passing per domain, with all 50,000 as the denominator.

Checks passing together	Domains	Share
0	46,252	92.50%
1	2,635	5.27%
2	280	0.56%
3	833	1.67%

Those 833 domains account for 91.54% of all MCP-marker matches. The Markdown check passes on 32.60% of domains passing llms.txt, compared with 0.85% of the remaining domains. Adding the three separate pass counts would count some domains more than once. After adjusting for the rank mix in the full million-domain list, the estimated share passing all three is 2.97%, and the share passing none is 91.91%.

4.2 Content access and request outcomes

For an AI agent to read a website without a full browser, two basic conditions must be met. First, the website must return readable text directly in its raw HTML without requiring browser JavaScript execution. Second, web security firewalls must not block automated requests.

In this study, the text-length check evaluates whether a website provides at least 300 characters of readable plain text directly on its homepage. This check passed on 22,855 domains (45.71%) and failed on 27,145 (54.29%), where pages were blank, required JavaScript rendering, or returned connection errors.

To evaluate whether web firewalls block automated agents, the scanner sent test requests identifying themselves as AI crawlers using the User-Agent headers for GPTBot (OpenAI) and PerplexityBot. Among domains passing the text check, both crawler-named requests returned HTTP 200 on 65.98%, one on 11.28%, and neither on 22.74%. The average checklist score in this group was 41.1.

Both conditions, readable text and unblocked access for both AI crawlers, occur on 15,080 domains, or 30.16% of the full sample. Requiring both conditions gives fewer matches than checking either alone. Because the text check only counts total characters rather than understanding meaning, a website returning a verbose 500-word error screen (such as an access-denied page) will pass the check, whereas a working website with a concise homepage under 300 characters will fail it.

Domains failing the text check still produced positive results on other checks. For example, 452 passed the llms.txt check and 2,231 passed the sitemap-and-robots check. Table 5 reports counts and percentages within each group; a domain can appear in several rows.

Table 5: Other check results conditional on the text-length label.

Check	Text pass (n=22,855)	Text fail (n=27,145)
HTTP 200 for both test requests	15,080 (65.98%)	3,832 (14.12%)
llms.txt check	2,858 (12.50%)	452 (1.67%)
Markdown-like response	1,452 (6.35%)	22 (0.08%)
Sitemap and robots.txt markers	11,455 (50.12%)	2,231 (8.22%)
JSON-LD script check	10,446 (45.71%)	241 (0.89%)
MCP response marker	890 (3.89%)	20 (0.07%)

The gap for the two-request check is 51.86 percentage points. Domains passing the text rule pass other checks much more often, so results from that group alone cannot describe the full sample. Some links between the checks also follow from how they work. The text and JSON-LD checks read the same selected page content, and the text check can use the Markdown reply if the ordinary reply is empty.

Keeping only domains that pass the text check would discard 13.66% of the llms.txt passes, 16.30% of the sitemap-and-robots passes, and 20.26% of the both-request HTTP-200 passes. Even when the text check fails, other addresses or request types can still produce positive labels. All domains remain in the main analysis.

4.3 Readiness profiles by rank

Final average scores differ by less than one point across the three rank groups. Before the low-score rechecks, the averages were 20.7, 22.4, and 23.9. The rechecks covered 52.44%, 43.12%, and 38.40% of the groups, respectively.

Table 6: Stored scores and request outcomes within each rank group.

Rank group	Domains	Average score	Median	Neither test returned HTTP 200
1-10,000	10,000	26.1	23	56.25%
10,001-100,000	20,000	25.2	20	55.58%
100,001-1,000,000	20,000	25.9	22	52.23%

Similar averages hide differences in the individual checks. Markdown and MCP-marker rates rise from the highest- to the lowest-ranked group, while the llms.txt rate is lower in the middle group. The basic MCP counts are six, 219, and 685. Two groups can have similar overall scores but different kinds of positive results.

Table 7: Selected pass rates within each rank group.

Rank group	llms.txt check	Markdown-like response	MCP response marker
1-10,000	6.38%	1.66%	0.06%
10,001-100,000	5.86%	2.48%	1.10%
100,001-1,000,000	7.51%	4.07%	3.43%

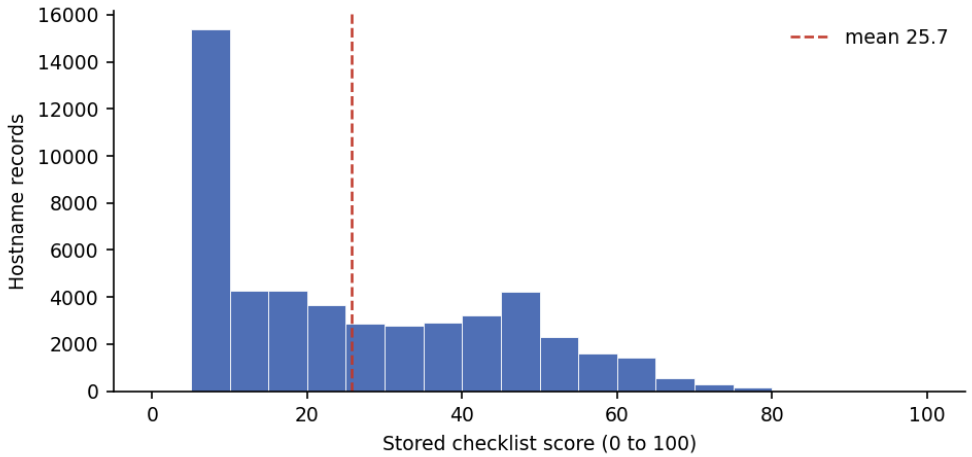


Figure 2: Number of domains in each score range, with the average marked.

4.3.1 Adjusting for the mix of ranks

In the 50,000 domain sample, top-ranked websites were intentionally overrepresented (20% of the sample, compared to only 1% in the full top-million list) to ensure high statistical confidence among the web's most prominent sites. To estimate what the results would look like across the entire top one million websites, the study applies rank-based weighting, giving each group its true proportional share (1% top tier, 9% middle tier, and 90% lower tier).

Table 8 compares the raw sample values against these population-adjusted estimates. The brackets show approximate 95% sampling confidence intervals.

Table 8: Sample values and rank-adjusted estimates for both record versions. Brackets give individual sampling-only 95% ranges.

Measure	Final sample	Earlier, adjusted	Final, adjusted
Checklist score	25.69	23.77 [23.53, 24.00]	25.87 [25.64, 26.10]
Neither test returned HTTP 200	54.37%	57.57% [56.96%, 58.18%]	52.57% [51.95%, 53.19%]
Text-length check	45.71%	41.41% [40.79%, 42.02%]	46.58% [45.96%, 47.20%]
HTTP 200 for both test requests	37.82%	33.88% [33.29%, 34.46%]	38.11% [37.51%, 38.71%]
Text check and both HTTP-200 replies	30.16%	26.60% [26.05%, 27.15%]	29.91% [29.35%, 30.48%]
Sitemap and robots.txt markers	27.37%	24.57% [24.03%, 25.10%]	27.54% [26.98%, 28.09%]
JSON-LD script check	21.37%	18.44% [17.96%, 18.92%]	20.67% [20.16%, 21.17%]
llms.txt check	6.62%	6.60% [6.29%, 6.91%]	7.35% [7.02%, 7.67%]
Markdown-like response	2.95%	3.54% [3.31%, 3.77%]	3.90% [3.65%, 4.14%]
MCP response marker	1.82%	2.91% [2.69%, 3.12%]	3.18% [2.96%, 3.41%]
OpenAPI version marker	0.16%	0.09% [0.05%, 0.13%]	0.10% [0.06%, 0.13%]
At least one of three selected checks	7.50%	7.26% [6.94%, 7.59%]	8.09% [7.75%, 8.43%]
Payment-related markers	0.33%	0.29% [0.22%, 0.36%]	0.33% [0.26%, 0.40%]

Weighting changes some indicators noticeably. Because MCP interface markers are more common among smaller websites in the lower rank group (3.43% versus 0.06% in the top tier), adjusting for the true rank mix raises the estimated MCP adoption rate from 1.82% to 3.18%. Conversely, candidate API descriptions with an OpenAPI version marker drop from 0.16% to 0.10%.

The joint text-and-both-request estimate is 29.91%, compared with 30.16% in the sample. The share passing at least one selected check rises from 7.50% to 8.09%, while the OpenAPI-marker estimate falls from 0.16% to 0.10%. Across both weighting models, traditional web discovery mechanisms (text, sitemaps, and JSON-LD) consistently outnumber specialized AI guidance files and agent protocols.

4.4 Comparison with the CrUX list

The Chrome User Experience Report (CrUX) is a public dataset published by Google that measures real-world user traffic and performance for popular web origins [2; 3]. Comparing Tranco domains against CrUX tests whether websites with verified human traffic demonstrate higher readiness for automated agents.

Among the sample, 23,592 domains appear in CrUX. These domains passed each tested check substantially more often than unlisted domains. The first two data columns in Table 9 show observed pass rates, while the final column shows the percentage point (pp) gap after standardizing both groups to the same rank distribution.

Table 9: CrUX comparisons before and after giving both groups the sample's rank mix.

Check	In CrUX (n=23,592)	Outside CrUX (n=26,408)	Gap with same rank mix (pp)
llms.txt check	8.86% (2,091)	4.62% (1,219)	+4.66
MCP response marker	2.83% (668)	0.92% (242)	+2.54
HTTP 200 for both test requests	45.78% (10,800)	30.72% (8,112)	+15.64
Markdown-like response	4.02% (948)	1.99% (526)	+2.50
robots.txt rule check	54.57% (12,875)	34.10% (9,006)	+21.22

Domains with verified user traffic in CrUX show higher pass rates across every rank group. For the bot-request check, the gap is 15.06 percentage points before adjustment and 15.64 after it. This indicates that sites with active human audiences are significantly more reachable by automated tools, though higher traffic alone does not explain all technical differences.

4.5 Where the checks can mislead

Automated checks rely on simple heuristics, which can sometimes produce misleading results when servers behave unexpectedly. For example, returning the same generic HTML page for every requested address produced 23 passes and an artificial score of 47. A verbose HTTP 403 ("Forbidden") error page passed the text check because it contained over 300 characters of error text, while a working static page with concise text failed it.

Table 10: Selected prepared-response tests and their recorded outcomes.

Test response	Result
Every request raises an error	All 61 labels are produced; score 6
Long HTTP 403 error page	Text check passes
Short static HTTP 200 page	Text check fails
Same generic HTML at every path	23 passes; score 47
HTTP 200 error reply containing jsonrpc	Basic MCP check passes
Capabilities reply wrapped as a server-sent event	Basic MCP check passes; capabilities check warns
Wildcard robots rule disallowing all paths	robots.txt rule check passes

Similarly, an empty HTTP 404 ("Not Found") response and a network connection timeout produced identical label profiles in the scanner. In the real dataset, 14,776 domain records (29.55%) match this blank profile, where the saved data cannot distinguish between a server that does not exist and one that blocked the connection.

Parsing rules can also create false negatives. Out of 910 basic MCP matches, only 3 passed the secondary capabilities check, while 907 received a warning. This occurred because the scanner expected a single JSON document, whereas many real MCP implementations stream their data using Server-Sent Events (SSE) [30]. The scanner failed to parse the stream even though the server was operating correctly. A warning therefore does not necessarily mean a service is invalid.

4.6 How repeat scans affect the comparison

In large-scale web measurement, transient network hiccups, temporary server throttling, or DNS timeouts can artificially lower scores on a first attempt. To evaluate this effect, the scanner rechecked 21,547 domains whose initial score was 10 or below.

These repeat scans raised the average score across all 50,000 domains from 22.68 to 25.69, a change of 3.00 points across the full sample. Among the rechecked domains, scores increased on 5,965, decreased on 553, and remained unchanged on 15,029.

Table 11: Paired pass gains and losses under the selective recheck policy.

Check	Pass gains	Pass losses	Net change (pp)
Text-length check	3,629	0	+7.26
HTTP 200 for both test requests	3,018	16	+6.00
llms.txt check	467	5	+0.92
Markdown-like response	171	0	+0.34
MCP response marker	84	0	+0.17

The text-check count rose from 19,226 (38.45%) to 22,855 (45.71%). Because only low-scoring domains were rechecked, almost no domains that originally passed had an opportunity to register a failure. The highest-ranked group gains 5.37 score points, compared with 1.99 in the lowest-ranked group. The repeat scans covered a larger share of the highest-ranked group, and the average gain among its rechecked domains was also larger, 10.24 versus 5.19 points. This shows that retry policies significantly influence measurement results and must be accounted for when comparing across studies.

5 What the Results Mean for Agent Readiness

The results reveal a sharp split across different web capabilities. Basic text, sitemaps, and JSON-LD metadata are relatively common, whereas dedicated agent guidance files and structured interfaces are rare. These numbers reflect specific matching rules rather than the percentage of sites that could support an agent. Table 12 summarizes the evidence across three core activities.

Table 12: Recorded sample evidence by activity and its interpretation.

Readiness area	Recorded evidence	Assessment
Content access	Text check 45.71%; text check and both "OK" replies 30.16%	Matches to two recorded conditions; content usefulness was not tested
Resource discovery	Sitemap-and-robots check 27.37%; JSON-LD 21.37%; llms.txt 6.62%	Common web formats pass more often than the dedicated guidance-file check
Service interaction	MCP marker 1.82%; OpenAPI marker 0.16%; payment markers 0.33%	Few matches at the tested addresses; successful actions were not tested

5.1 Reading and discovering information

Because websites format information differently, automated agents benefit from exploring multiple discovery paths, such as standard pages, XML sitemaps, structured JSON-LD data, or llms.txt files.

Most MCP server detections occur on domains that also provide llms.txt and Markdown responses. This clustering suggests that early adopters often deploy multiple AI standards at once. However, checking homepage text alone is not a reliable filter for whether a site provides machine resources. A domain with minimal homepage text can still host working APIs at other addresses, while a website serving a verbose error page can mistakenly pass the text check simply because the error message contains hundreds of words. Each resource requires independent inspection.

5.2 Moving from information to action

The low match rates for MCP (1.82%) and OpenAPI (0.16%) reflect the limits of probing fixed default paths without client-side script rendering. A service may host an API at an unprobed address or require authentication before revealing its endpoints.

Furthermore, detecting an endpoint marker does not prove that an agent can execute an action. Server responses wrapped in Server-Sent Events can cause simple parser checks to fail even when the underlying service is fully functional. Positive and negative checks indicate whether responses matched specific heuristic patterns, not whether an agent could complete a live task.

5.3 Practical priorities for websites and agents

For website operators wishing to support autonomous software, ensuring that pages return clean text without mandatory JavaScript rendering, publishing XML sitemaps, and providing clear API documentation offer the most immediate benefits.

For agent developers, verifying file contents is essential. As shown by Thunderbit, many HTTP 200 responses to llms.txt paths return regular web pages rather than markdown guidance. Agents must verify response structure rather than assuming success from HTTP status codes alone.

A rigorous next step would be evaluating agent task success across domains that pass versus fail these infrastructure checks, testing whether these static indicators predict practical task completion.

6 Limitations

This study evaluates recorded check results rather than complete task executions. The saved records store categorical labels (pass, warning, fail, not applicable) rather than full HTTP response bodies, exact error traces, or redirect chains. This prevents retrospectively testing alternative text thresholds or diagnosing network timeout causes.

The checklist weights various indicators, but individual checks do not matter equally for every task. A research agent only requires readable text and sitemaps, while a transaction agent requires APIs and payment flows. The study does not establish that higher scores increase task completion rates.

Additionally, measurements reflect the specific request headers, paths, timeouts, and single-attempt retry rules used during collection. Crawler test requests used identifiable User-Agent strings but do not reflect full crawler behavior. Finally, private corporate intranets, authenticated services, and pages requiring interactive browser rendering lie entirely outside this study's scope.

7 Conclusion

This study measures the public web's readiness for autonomous AI agents across 50,000 domains sampled from the Tranco top one million. The text check passes on 45.71% of domains. Across all domains, 30.16% pass that check and receive HTTP 200 for both crawler-named requests. At least one of the three checks for llms.txt, Markdown, and a basic MCP marker passes on 7.50% of domains. These are recorded check results, not a validated readiness grade for the web.

Dedicated machine interfaces remain rare on the public web, with basic MCP and OpenAPI markers on 1.82% and 0.16%, respectively, at the tested addresses. The study did not execute complete tasks. Its practical contribution is a reproducible baseline of access, discovery, and interface readiness across the global web, showing that while agents can read static information from many sites, structured public infrastructure for automated service interaction remains rare.

Appendix A. Complete Check Results

Each row sums to 50,000. Appendix B explains the labels. NA means not applicable and includes optional checks and checks that depend on another result.

Check	Pass	Warning	Fail	NA
access-boilerplate-ratio	7,737	42,263	0	0
access-compression	28,210	21,790	0	0
access-js-hydration	22,855	0	27,145	0
agent-card	3,791	46,209	0	0
agent-payments	163	0	0	49,837
agents-json	3,828	46,172	0	0
agents-txt	4,136	45,864	0	0
ai-catalog	3,890	46,110	0	0
api-catalog-rfc9727	3,883	46,117	0	0
api-cors-ai	3,215	46,785	0	0

Check	Pass	Warning	Fail	NA
api-dry-run	13	5,559	0	44,428
ard-catalog	20	3,566	46,414	0
auth-guide	4,372	45,628	0	0
bot-ua-access	18,912	3,901	27,187	0
canonical	15,771	0	34,229	0
csp	9,661	40,339	0	0
favicon-branding	26,086	23,914	0	0
frame-protection	16,580	33,420	0	0
hsts	14,243	35,757	0	0
html-lang	25,119	24,881	0	0
https-tls	50,000	0	0	0
json-ld	10,687	0	39,313	0
link-headers	2,173	47,827	0	0
llms-full	5,771	44,229	0	0
llms-txt	3,310	5,418	41,272	0
markdown-negotiation	1,474	0	48,526	0
markdown-twins	5,337	44,663	0	0
mcp-schema-handshake	3	907	0	49,090
mcp-jsonrpc-screen	910	49,090	0	0
meta-description	13,634	5,657	30,709	0
oauth-agent-discovery	5,709	44,291	0	0
og-tags	14,486	0	35,514	0
openai-plugin	4,817	45,183	0	0
openapi-examples	50	5,522	0	44,428
openapi-spec	78	49,922	0	0
payments-terms	4,738	45,262	0	0
payments-webln	30	0	0	49,970
permissions-policy	4,962	45,038	0	0
rate-limit-headers	361	49,639	0	0
referrer-policy	10,313	39,687	0	0
rfc-7807-errors	6	0	0	49,994
robots-ai-policy	21,881	3,638	24,481	0
robots-meta-ai	46,390	0	3,610	0
robots-selective-ai-policy	4,420	21,099	24,481	0
rss-feed	3,365	46,635	0	0
security-ai-disclosure	23,382	26,618	0	0
security-c2pa	1	49,999	0	0
security-signatures	4,096	45,904	0	0
security-txt	2,288	47,712	0	0
seo-answer-first	15,310	34,690	0	0
seo-author-eeat	6,910	43,090	0	0
seo-freshness	3,632	46,368	0	0
seo-multimodal	47,814	2,186	0	0
seo-rich-schemas	6,230	43,770	0	0
seo-schema-graph	8,474	41,526	0	0
sitemap-json	2,903	47,097	0	0
sitemap-xml	13,686	3,581	32,733	0
tdmrep	3,514	46,486	0	0
title-tag	25,103	4,192	20,705	0
twitter-cards	11,871	38,129	0	0
xcto	14,596	35,404	0	0

APPENDIX B. CHECK DEFINITIONS

These tables explain exactly what produces a pass. GET asks for content and POST sends data. The response body is the content returned, while headers give information alongside it. HTTP 2xx means any status from 200 through 299. A nonempty body contains at least one character. JSON is a structured text format; parsing means reading its structure. A regex is a text-matching pattern.

The tables keep exact field names and paths so the rules can be reproduced. Each path starts at the domain's HTTPS address unless a reply gives another URL to try. The check named mcp-jsonrpc-screen is stored as mcp-server-live in the data. The interface category is stored as usability.

B.1 Discovery

The discovery checks look for crawler rules [4], sitemaps [5], guidance for language models [6], and API lists [7]. ARD is the program's chosen address for a possible Agent Readiness Directory. The llms.txt check requires more than the proposal's minimum project heading and tries only the listed addresses.

Check	Request	Match rule
robots-ai-policy	GET /robots.txt	Nonempty HTTP-2xx body; no regex match for root-wide Disallow after GPTBot, or no such match after ClaudeBot
robots-selective-ai-policy	GET /robots.txt	Nonempty HTTP-2xx body contains a token from each of two scanner-defined bot-name lists
llms-txt	GET /llms.txt; try /.well-known/llms.txt after failure, non-2xx, or empty body	HTTP-2xx body over 50 chars with an H1-style line and Markdown link syntax
llms-full	GET /llms-full.txt and /llms-small.txt	HTTP-2xx full-file body over 100 chars, or any nonempty HTTP-2xx small-file body
ard-catalog	GET /.well-known/ard.json	Nonempty HTTP-2xx body parses as JSON
ai-catalog	GET /.well-known/ai-catalog.json	Any HTTP-2xx response
api-catalog-rfc9727	GET /.well-known/api-catalog	Nonempty HTTP-2xx body
sitemap-xml	GET /sitemap.xml and a robots-declared URL; /sitemap_index.xml after root-sitemap failure or empty body	Selected HTTP-2xx body contains <urlset or <sitemapindex; robots text also has a Sitemap HTTP-URL directive
sitemap-json	GET /.well-known/sitemap.json	Nonempty HTTP-2xx body contains {
agents-txt	GET /agents.txt	HTTP-2xx body over 20 chars
agents-json	GET /.well-known/agents.json	Any HTTP-2xx response
agent-card	GET /.well-known/agent-card.json	Any HTTP-2xx response [9]
openai-plugin	GET /.well-known/ai-plugin.json and /plugin.json	Any HTTP-2xx response at either path

B.2 Access

These checks inspect response codes, returned text, and headers. They also ask the server for particular content formats, a process called content negotiation [13], and examine the Link header [14].

Check	Request	Match rule
markdown-negotiation	GET / with Accept: text/markdown, text/plain;q=0.9	HTTP-2xx; Content-Type contains text/markdown or the body matches the H1/front-matter regex
markdown-twins	GET /index.md	HTTP-2xx body over 50 chars
bot-ua-access	GET / with GPTBot and PerplexityBot user-agent strings	Both requests classified as HTTP 200
robots-meta-ai	Selected meta robots, meta googlebot, and X-Robots-Tag	No noindex or nosnippet substring; extra points for specified snippet or image-preview markers
link-headers	Selected Link header	Exact double-quoted rel value describedby, service-desc, or alternate
rate-limit-headers	Selected rate headers	Nonempty ratelimit-limit, x-ratelimit-limit, ratelimit-remaining, or Retry-After value
access-js-hydration	Selected root body, or Markdown-request body as fallback	More than 300 characters after script, style, tag, and whitespace processing
access-compression	Selected content-encoding header	Lowercased value contains br, zstd, or gzip

Check	Request	Match rule
access-boilerplate-ratio	Extracted text length divided by selected body length	Rounded percentage at or above 10%

B.3 Interfaces

These checks look for a reply to an MCP connection request [8], an OpenAPI description of an API [10], OAuth information about access permissions [11; 12], and a TDMRep file about text-and-data-mining preferences [20]. Some rules below use JavaScript's idea of a value being true, called "truthy". For example, a nonempty string or an object counts as true, while a missing value, an empty string, or false does not.

Check	Request	Match rule
mcp-jsonrpc-screen	POST initialize to /.well-known/mcp and /api/mcp	Either HTTP-2xx body contains jsonrpc
mcp-schema-handshake	After a basic MCP pass, parse the first nonempty body in well-known then API-path order	JSON parses; Boolean(result.capabilities) is true
openapi-spec	GET /openapi.json and /.well-known/openapi.json; try /openapi.yaml if neither fetch is HTTP-2xx; select first qualifying body over 50 chars	Parsed JSON has a truthy openapi or swagger value; on parse failure, body contains openapi: or swagger:
openapi-examples	Selected OpenAPI-candidate text	Case-insensitive match for "example": or "examples":
auth-guide	GET /auth.md	HTTP-2xx body over 30 chars
oauth-agent-discovery	GET /.well-known/oauth-authorization-server and /.well-known/oauth-protected-resource	Any HTTP-2xx response at either path
tdmrep	GET /.well-known/tdmrep.json	Nonempty HTTP-2xx body
rfc-7807-errors	GET /api	Content-Type contains application/problem+json
api-dry-run	Selected OpenAPI-candidate text	Case-insensitive quoted dry_run, dryRun, validate_only, idempotency_key, or Idempotency-Key token
api-cors-ai	First truthy Access-Control-Allow-Origin value in root, /api, then /openapi.json order	Selected value remains nonempty after trimming

B.4 Security

This group looks for security-related headers such as HSTS [15] and Content Security Policy [16], and security contact information in security.txt [17]. The HTTPS check only reads the start of the input URL. It therefore passes for every domain in this sample without testing the connection itself.

Check	Request	Match rule
https-tls	Normalized input scheme	Input begins with HTTPS
hsts	Selected Strict-Transport-Security	Contains max-age
csp	Selected Content-Security-Policy	Value longer than 10 chars
xcto	Selected X-Content-Type-Options	Lowercased value contains nosniff
frame-protection	Selected headers	Nonempty X-Frame-Options, or CSP contains frame-ancestors
referrer-policy	Selected Referrer-Policy	Nonempty value
permissions-policy	Selected Permissions-Policy	Nonempty value
security-txt	GET /.well-known/security.txt; try /security.txt after failure, non-2xx, or empty body	HTTP-2xx body contains Contact:

Check	Request	Match rule
security-signatures	GET /.well-known/http-message-signatures-directory	Any HTTP-2xx response
security-c2pa	Selected root-body text	Contains application/c2pa or c2pa.org
security-ai-disclosure	Bodies from /.well-known/security.txt and /terms-of-use.md	Regardless of HTTP status, case-insensitive AI/LLM/Agent/Prompt-Injection or AI/Safety/Model substrings, respectively

B.5 METADATA

Metadata describes a page for other software. These checks include content labelled as JSON-LD [18] and terms associated with Schema.org [19].

Check	Request	Match rule
title-tag	Regex-selected title-tag text	Trimmed length from 10 through 75 chars
meta-description	Selected meta-description text	Length from 50 through 170 chars
canonical	Selected canonical-link value	Starts with http://, https://, or /
html-lang	Selected html lang attribute	Trimmed value has at least two chars
favicon-branding	GET /favicon.ico and selected root body	HTTP 200 from icon path, or a matching icon-link marker
og-tags	Selected Open Graph values	Nonempty og:title and either og:description or og:image
twitter-cards	Selected Twitter values	Nonempty twitter:card or twitter:title
json-ld	Script-content regex with an application/ld+json marker	At least one selected block parses as JSON
seo-schema-graph	Selected root-body text	One of "@graph", "isPartOf", "mainEntity", or "author"
seo-rich-schemas	Selected root-body text	Case-insensitive quoted FAQPage, HowTo, Article, BlogPosting, SoftwareApplication, WebApplication, Course, DefinedTermSet, or BreadcrumbList
seo-author-eeat	Selected root-body text	Contains "sameAs", wikidata.org, orcid.org, scholar.google.com, or both "Person" and "name"
seo-answer-first	Selected root-body text	H1 and H2 tag regexes both match
seo-multimodal	Selected root-body text	No lowercase img opening marker, or at least one image matches nonempty alt text
seo-freshness	Selected root-body text	Case-insensitive last_updated, lastmod, published_time, or nonempty quoted dateModified/datePublished value
rss-feed	GET /feed.xml and /feed.json plus a root-body link URL when found; prefer qualifying XML, then any HTTP-2xx JSON response, then the declared response	Selected body contains <rss, <feed, "version", or the exact quoted Atom namespace marker

B.6 PAYMENTS

Check	Request	Match rule
agent-payments	Selected headers and root-body text	L402/LSAT authentication marker, nonempty x-payment header, or x402/lightning: text
payments-webln	Selected root-body text	Contains webln or lightning:
payments-terms	GET /terms-of-use.md and selected root body	Any HTTP-2xx from terms path, or terms-of-use.md or ai-terms text

B.7 RESPONSE SELECTION

Text checks use the ordinary homepage reply if its body contains any characters. Otherwise, they use the Markdown reply, or empty text if neither has content. Several header checks use the first

available header set in this order, ordinary homepage, Markdown request, then GPTBot request. The program makes these choices without requiring a successful response code. Text and header checks can therefore read different replies.

The text rule removes scripts, styles, and tags using text patterns, then reduces repeated spaces and line breaks. It uses JavaScript's string length, measured in UTF-16 code units, so some characters count as two. Hidden text also counts, and a written HTML code such as & stays encoded rather than becoming one displayed character. Failed requests give the checks no response object, and failures while reading a reply give them empty text.

B.8 Request settings

The rules follow scanner revision ea0e797. That version predates the saved results and uses the same check names and record format.

Most requests use VedaAgenticScanner/1.0 (+https://veda.ng/scan; agent-readiness-audit). The Markdown root request adds Accept: text/markdown, text/plain;q=0.9. The two bot root requests use Mozilla/5.0 (compatible; GPTBot/1.0; +https://openai.com/gptbot) and Mozilla/5.0 (compatible; PerplexityBot/1.0; +https://perplexity.ai/perplexitybot).

MCP requests use POST and Content-Type: application/json. Their JSON-RPC 2.0 initialize payloads specify protocol version 2024-11-05, empty capabilities, and client name veda-scanner version 1.0. Request IDs are probe-1 and probe-2. The implementation stops after the response and lacks the Accept-header and event-stream handling required by the later Streamable HTTP transport [30].

The program follows redirects. It does not separately try adding www or changing HTTPS to HTTP. A sitemap or feed link can lead to another host, but the result still counts under the starting domain. Requests do not log in. The program fetches robots.txt alongside the other resources instead of checking its rules before making requests. It does not submit forms, make payments, or keep cookies between requests. The watchdog allows three domains to be processed at once in each process and can run several processes.

Appendix C. Scores and Sampling Weights

C.1 Checklist score

The program compares the points earned with the maximum points from checks it marks neither optional nor not applicable. It expresses the result as a percentage, rounds to a whole number, and caps it at 100. Optional checks can add points without increasing the maximum used in this calculation. If that maximum is zero, the program uses one instead.

Two results with the same label can earn different points, so this paper uses the scores saved by the program. The input-HTTPS rule adds three points per domain. Category scores use the same method. Their averages are 41.0 for security, 28.9 for metadata, 23.9 for access, 20.7 for discovery, 9.8 for payments, and 7.4 for interfaces.

Optional points can increase the score without increasing its denominator, while applicability changes which checks enter that denominator. Equal scores therefore need not represent equal sets of passed checks or equal capabilities. The score comparisons describe the retained scoring system; the individual labels carry the more specific observations.

C.2 Why the results are weighted

The sample gives highly ranked domains a larger share than they have in the full Tranco list [1]. Weighting means giving each rank group its share of that full list when combining the results. The top 10,000 contribute 1%, the next 90,000 contribute 9%, and the remaining 900,000 contribute 90%. These figures estimate outcomes under the same collection procedure for the full list, assuming comparable measurement conditions. They do not estimate all websites or remove the effect of selective rechecking. The separate CrUX comparison in Section 3.5 gives listed and unlisted domains the same chosen mix of ranks.

The bracketed 95% ranges use a normal approximation with a correction for the share of each rank group sampled. They describe uncertainty from sampling only some domains in the lower two groups. All top-10,000 domains were included, so there is no sampling uncertainty for that group. These are separate ranges for each measure, conditional on the saved labels. They do not include request variability, mistaken matches, or uncertainty about repeat scans, and do not provide a combined test across the checklist. The analysis code gives the detailed calculations.

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